



Shape and operator embeddings of sequential data for machine learning

Thibaut Germain

CMAP, Ecole Polytechnique, Institut Polytechnique de Paris

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MALT seminar

Shape-based embeddings

How to learn time series embeddings that account for their shape ?

Shape analysis for time series

Germain, T., Gruffaz, S., Truong, C., Durmus, A., & Oudre, L.

Advances in neural information processing systems (NeurIPS), 2024.

Motivation

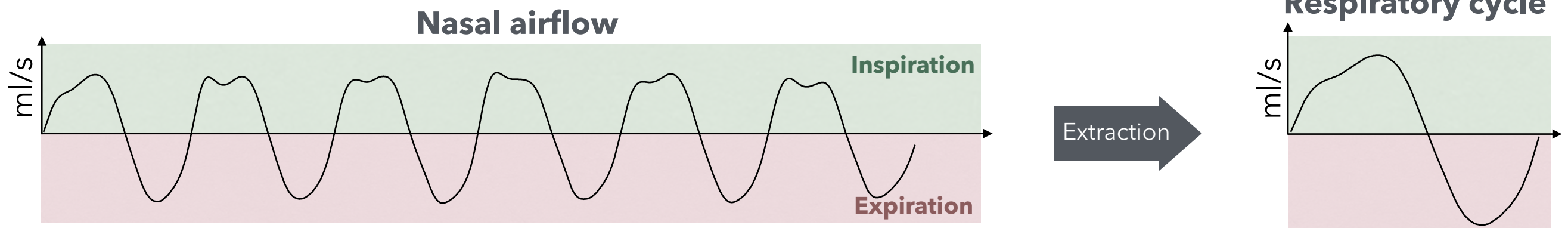
Inspired by a use case: mouse ventilation analysis

Aim: Assessing ventilation response after the exposure of a neurotoxin for mice of different genotypes.

Experimental protocol: For each genotype (WT, ColQ):



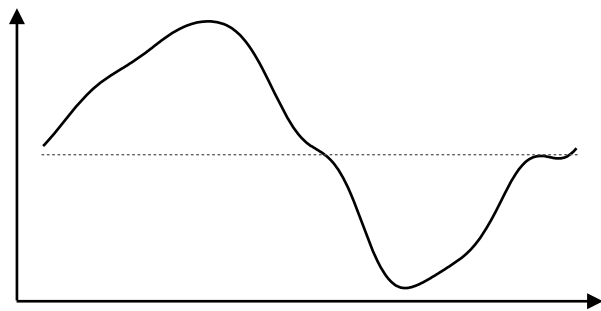
First insight: Time series with elementary patterns:



Inspired by a use case: mouse ventilation analysis

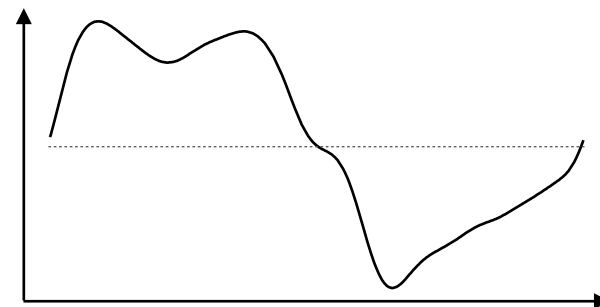
Biologist's insights: Shapes of respiratory cycles accounts for physiological states.

WT (before)



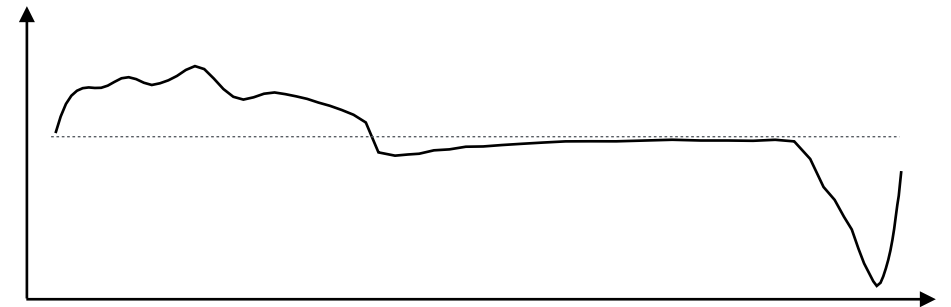
« Healthy mouse's respiratory cycles »

ColQ (before)



« The two bumps during inspiration indicate fine-motor control difficulties »

WT (after)



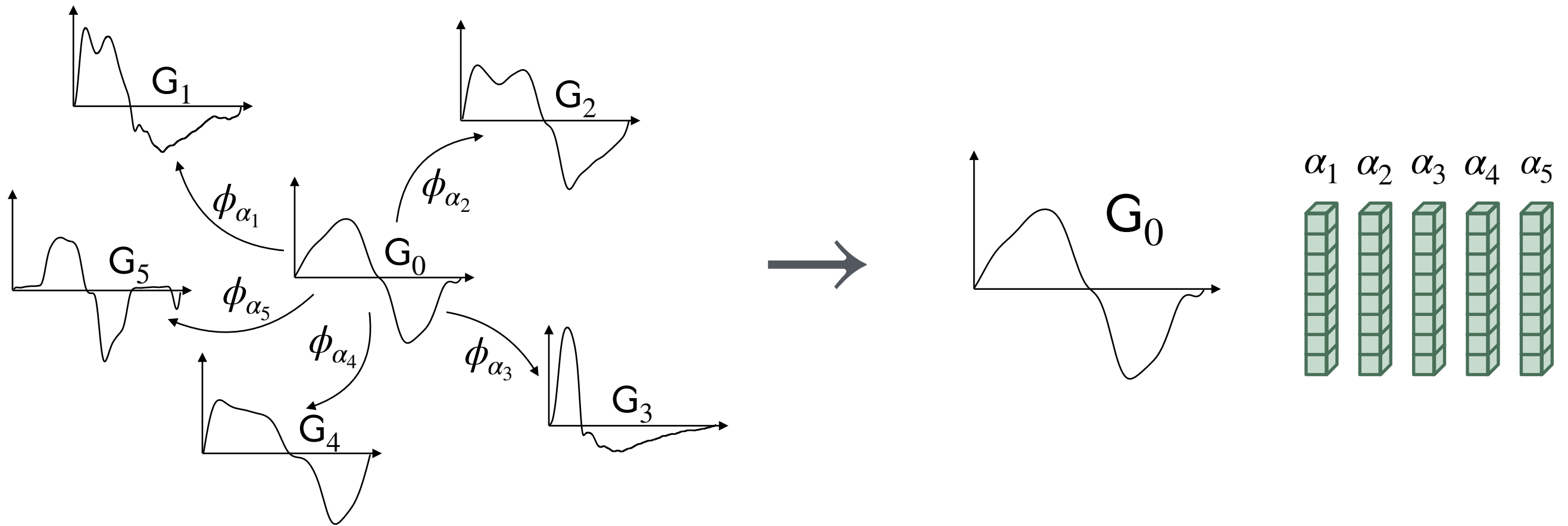
- « The constrained flow during inspiration indicates airways obstructions »
- « The long pause indicates tetany, its duration is important »

Transcription to modeling & requirements:

- Embed time series as **deformations of a referent time series**
- **Two-way mapping** between temporal and embedding spaces
- **Compatible with statistical analysis** (identifiable model on a vector space)
- Work for **small datasets** with **irregularly sampled** and **variable-length** time series

Embedding framework outlines

Embedding outlines



Input: $\left(s_j : \mathbf{I}_j \mapsto \mathbb{R}^d\right)_{j \in [N]}$

1. The shape of s_j is $\mathbf{G}_j = \{(t, s_j(t)) : t \in \mathbf{I}_j\}$.
2. $\mathbf{G}_j$ is represented as the deformation of a reference graph $\mathbf{G}_0 = \{(t, s_0(t)) : t \in \mathbf{I}\}$, i.e.:

$$\mathbf{G}_j \approx \phi_{\alpha_j} \cdot \mathbf{G}_0 = \{\phi_{\alpha_j}(t, s_0(t)) : t \in \mathbf{I}\},$$
 where $\phi_{\alpha_j} : \mathbb{R}^{d+1} \mapsto \mathbb{R}^{d+1}$ is a diffeomorphism parametrized by $\alpha_j \in \mathbb{R}^m$.

Output: a graph of reference $\mathbf{G}_0 \subset \mathbb{R}^{d+1}$, deformation parameters $(\alpha_j)_{j \in [N]} \in (\mathbb{R}^m)^N$

How to learn G_0 and α_j 's ?

An optimization problem: $G_0 \subset \mathbb{R}^{d+1}$ and $(\alpha_j)_{j \in [N]} \in (\mathbb{R}^m)^N$ are learned by solving the empirical Fréchet mean problem:

$$\arg \min_{G_0, (\alpha_j)_{j \in [N]}} \frac{1}{N} \sum_{j \in [N]} \left(\underbrace{d_G^2(\phi_{\alpha_j} \cdot G_0, G_j)}_{\text{data attachment}} + \underbrace{\lambda d_\Phi^2(\text{Id}, \phi_{\alpha_j})}_{\text{regularization}} \right)$$

Open questions:

1. How to parametrized the diffeomorphisms $(\phi_{\alpha_j})_{j \in [N]}$?
2. What distances (d_G, d_Φ) should be considered ?

Remark: Description of d_G is not covered, intuitively it can be think as a maximum mean discrepancy (MMD) between graphs seen as measures [Kal17].

Generating diffeomorphisms: The LDDMM framework [Beg05]

How to parametrized diffeomorphisms ?

Theorem: Global diffeomorphism for small perturbation to the identity.

Let $f : \mathbb{R}^n \mapsto \mathbb{R}^n$ be a continuously differentiable (C^1) function such that:

$$\|Df(x)\|_\infty < 1, \quad \text{for any } x \in \mathbb{R}^n.$$

Then the map:

$$F(x) \triangleq x + f(x)$$

is a C^1 -diffeomorphism from $\mathbb{R}^n$ to $\mathbb{R}^n$.

Intuition: Diffeomorphism build by composition:

$$F = F_N \circ F_{N-1} \circ \dots \circ F_1$$

with $F_i = Id + f_i$ and $\|Df_i\|_\infty < 1$, for any $i \in [N]$.

Our case: Diffeomorphism defined by infinite composition, i.e. it is the flow of an ordinary differential equation (ODE).

Building diffeomorphisms with LDDMM [Beg05]

Theorem [Gla05]: Assuming $v \in L^2([0,1], V)$ a time-varying velocity field in $\mathbb{R}^n$, where V is an RKHS with some regularity assumptions [Gla05]. For any $x_0 \in \mathbb{R}^n$, the differential system:

$$\frac{dX(\tau)}{d\tau} = v_\tau(X(\tau)) \quad \text{with} \quad X(0) = x_0,$$

has a unique solution defined for all $\tau \in [0,1]$.

At any time $\tau \in [0,1]$, **the flow application:**

$$\phi_v^\tau : x_0 \in \mathbb{R}^n \mapsto x_0 + \int_0^\tau v_r(\phi_v^r(x_0)) dr \in \mathbb{R}^n$$

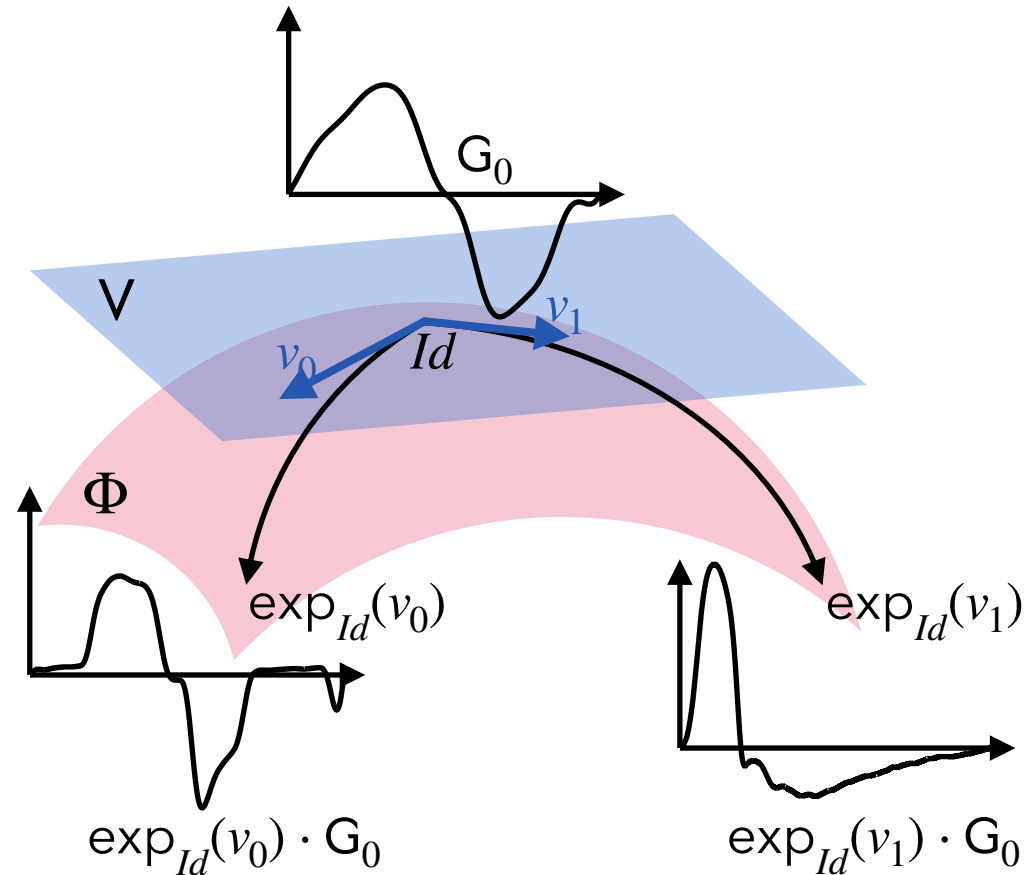
is a **diffeomorphism**.

A metric group: The group of diffeomorphisms $\Phi \triangleq \{\phi_v^1 \mid v \in L^2([0,1], V)\}$ is metrizable such that for any $\phi \in \Phi$:

$$d_\Phi^2(Id, \phi) = \inf_{v \in L^2([0,1], V)} \left\{ \int_0^1 \|v_\tau\|_V^2 d\tau \mid \phi_v^1 = \phi \right\},$$

the infimum is reached with a v^* and it conserves its norm along its geodesic path i.e.: $\|v_\tau^*\|_V = \|v_0^*\|_V, \forall \tau \in [0,1]$.

Geodesic shooting



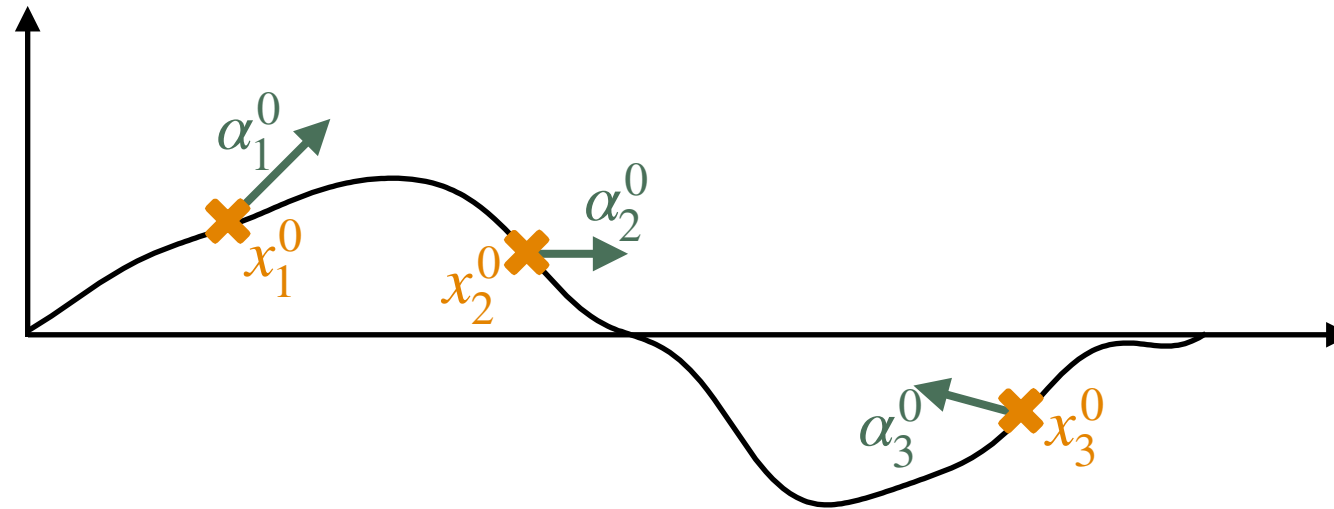
An exponential map: By denoting $\tau \mapsto \rho_{v_0}(\tau)$ the geodesic starting from Id with initial conditions $v_0 \in V$, the exponential map is [Mil06]:

$$\exp_{Id} : v_0 \in V \mapsto \rho_{v_0}(1) \in \Phi \quad \text{and} \quad d_{\Phi}^2(Id, \exp_{Id}(v_0)) = \|v_0\|_V^2$$

Important remarks:

- The exponential map is locally bijective.
- The geodesic path is uniquely define by the initial conditions.

Geodesic shooting in practice



Let denote $K : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^{n^2}$ the kernel of the RKHS $\mathcal{V}$.

Initial velocity field: Given N_0 control points $X_0 = (x_k^0)_{k \in [N_0]} \in (\mathbb{R}^n)^{N_0}$, and momentums $\alpha = (\alpha_k^0)_{k \in [N_0]} \in (\mathbb{R}^n)^{N_0}$, the initial velocity field is,

$$v_0 : x \in \mathbb{R}^n \mapsto \sum_{k \in [N_0]} K(x_k^0, x) \alpha_k^0 \in \mathbb{R}^n .$$

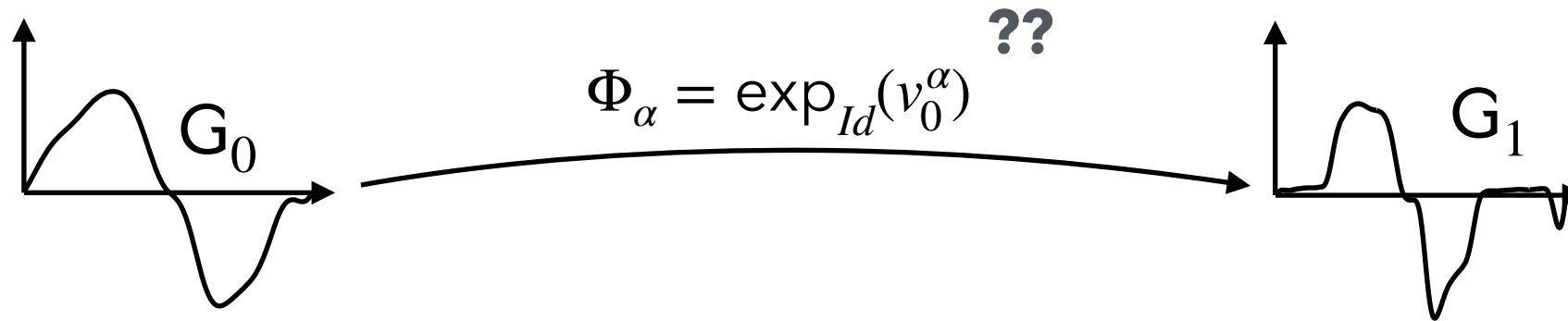
Along the geodesic path: For any $\tau \in [0,1]$, the velocity field verifies:

$$v_\tau : x \in \mathbb{R}^n \mapsto \sum_{k \in [N_0]} K(x_k(\tau), x) \alpha_k(\tau) \in \mathbb{R}^n ,$$

Geodesic equations:

$$\begin{cases} \frac{dx_k(\tau)}{d\tau} = v_\tau(x_k(\tau)) \\ \frac{d\alpha_k(\tau)}{d\tau} = - \sum_{l \in N_0} d_{x_k(\tau)} K(x_k(\tau), x_l(\tau)) \alpha_l(\tau)^\top \alpha_k(\tau) \end{cases} \quad \text{with} \quad \begin{cases} x_k(0) = x_k^0 \\ \alpha_k(0) = \alpha_k^0 \end{cases} \quad \forall k \in [N_0] .$$

Shape registration



Question: How to learn the initial momentum $\alpha \in (\mathbb{R}^n)^{N_0}$ encoding the diffeomorphism mapping G_0 to G_1 ?

Minimize: $\text{loss}(\alpha) \triangleq d_G^2(\exp_{Id}(v_0^\alpha) \cdot G_0, G_1) + \lambda \|v_0^\alpha\|_V^2$

Protocol: Given G_0 and G_1 do:

Set $\alpha = 0$

While did not converge:

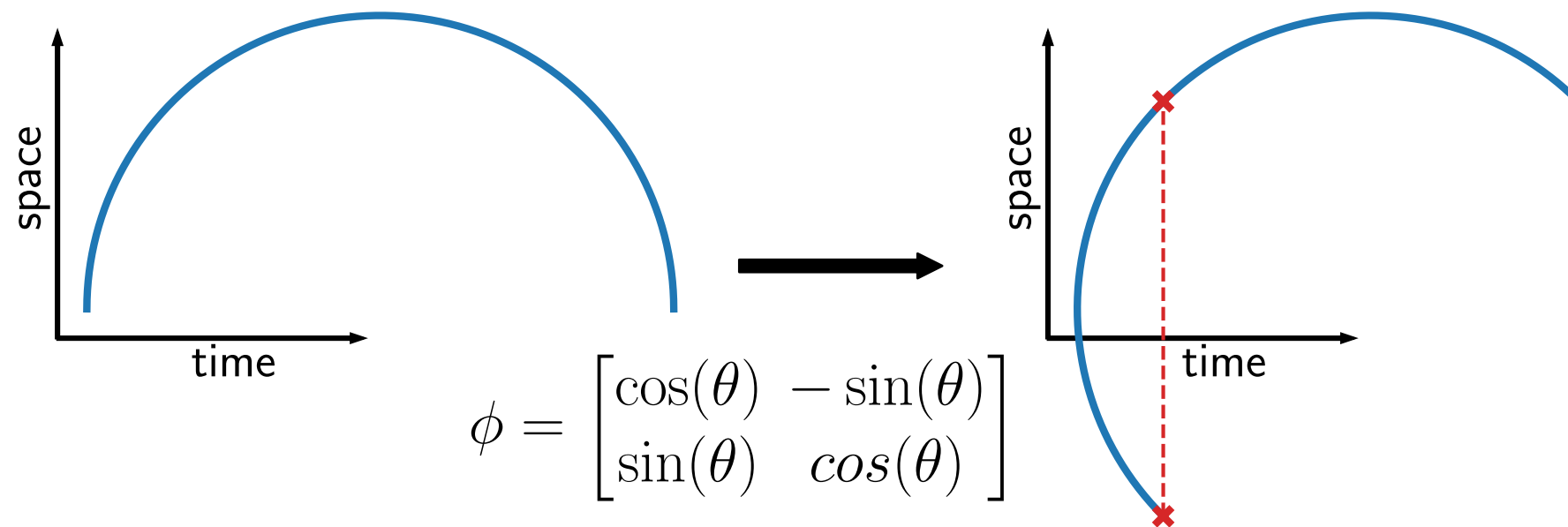
1. Perform geodesic shooting: $\exp_{Id}(v_0^\alpha)$ (ODE resolution: RK4)
2. Compute $\text{loss}(\alpha)$
3. Backward gradient (classical autodiff or reverse ODE with the adjoint method)

Return α

Application to time series

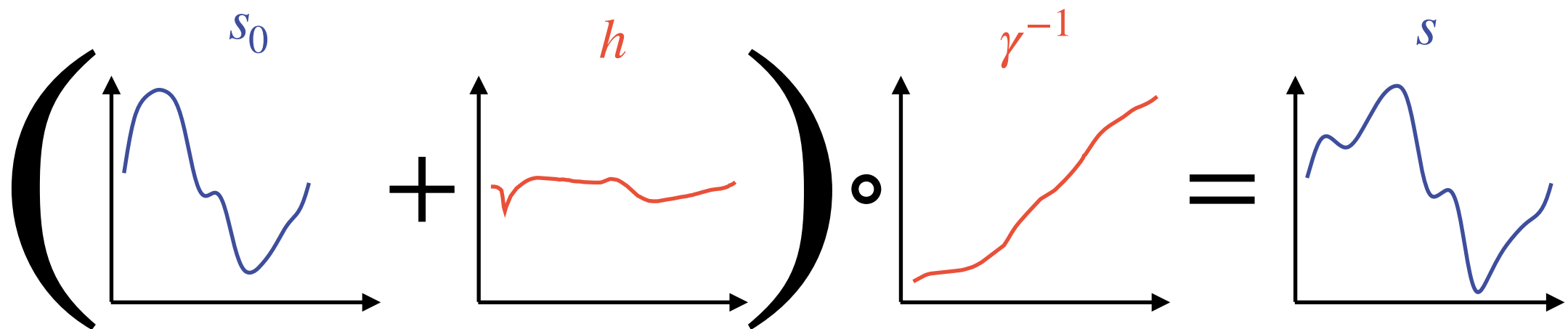
Respecting the graph structure of time series

Problem: A general diffeomorphism does not necessarily preserve the structure of a time series graph.



Deformation of times series

Intuition: Let $s_0 : \mathbf{I} \mapsto \mathbb{R}^d$ and $s : \mathbf{J} \mapsto \mathbb{R}^d$, the diffeomorphic deformation ϕ mapping s_0 to s should be seen as distortion $h : \mathbf{I} \mapsto \mathbb{R}$ and a time parametrization $\gamma^{-1} : \mathbf{J} \mapsto \mathbf{I}$ such that: $\phi \cdot \mathbf{G}(s_0) = \mathbf{G}((s_0 + h) \circ \gamma^{-1}) = \mathbf{G}(s)$.

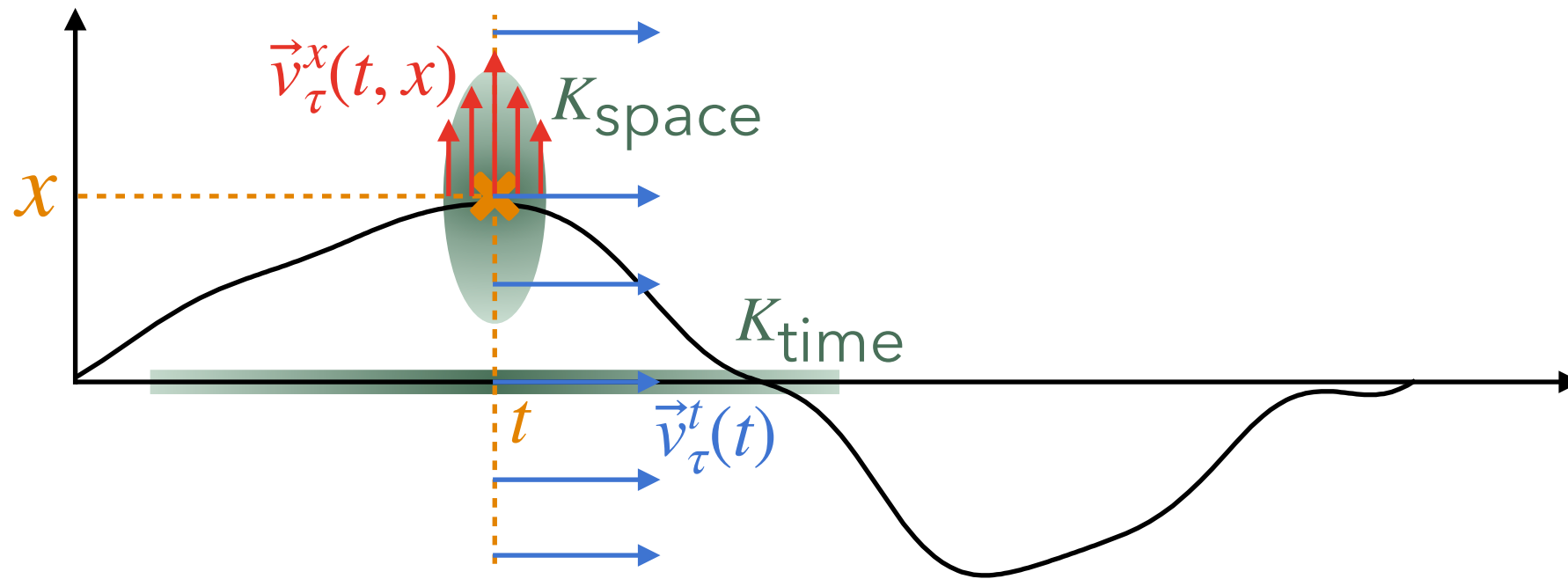


Theorem (representativity): Any continuously differentiable time series $s_0 : \mathbf{I} \mapsto \mathbb{R}^d$ and $s : \mathbf{J} \mapsto \mathbb{R}^d$, there exists deformations $\Psi_\gamma : (t, x) \in \mathbb{R}^{d+1} \mapsto (\gamma(t), x) \in \mathbb{R}^{d+1}$ with $\gamma \in \mathbf{D}(\mathbb{R})$, and $\Pi_f : (t, x) \in \mathbb{R}^{d+1} \mapsto (t, f(t, x)) \in \mathbb{R}^{d+1}$ with $f \in \mathbf{C}^1(\mathbb{R}^{d+1}, \mathbb{R}^d)$, such that:

$$\phi_{\gamma, f} \cdot \mathbf{G}(s_0) = \mathbf{G}(s) \quad \text{with} \quad \phi_{\gamma, f} = \Psi_\gamma \circ \Pi_f,$$

Moreover, for any $\bar{\gamma} \in \mathbf{D}(\mathbb{R})$, and $\bar{f} \in \mathbf{C}^1(\mathbb{R}^{d+1}, \mathbb{R}^d)$, $\phi_{\bar{\gamma}, \bar{f}} \cdot \mathbf{G}(s_0)$ is the graph of a continuously differentiable time series.

A kernel for time series



The proposed kernel: Let K_G be a kernel defined for any $(t, x), (t', x') \in (\mathbb{R}^{d+1})^2$,

$$K_G((t, x), (t', x')) = \begin{pmatrix} c_0 K_{\text{time}} & 0 \\ 0 & c_1 K_{\text{space}} \end{pmatrix} \quad \text{with} \quad \begin{cases} K_{\text{time}} = K_{\sigma_{T,0}}^{(1)}(t, t') \\ K_{\text{space}} = K_{\sigma_{T,1}}^{(1)}(t, t') K_{\sigma_x}^{(d)}(x, x') I_{\mathbb{R}^d} \end{cases}$$

parametrized by $\sigma_{T,0}, \sigma_{T,1}, \sigma_x > 0$ and the constants $c_0, c_1 > 0$.

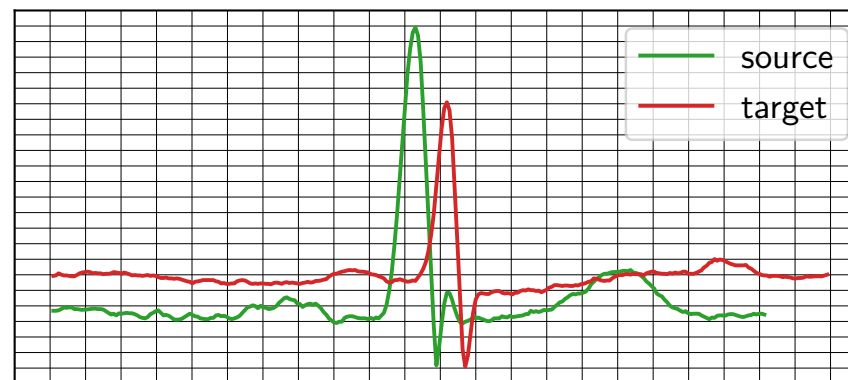
Lemma: For any initial velocity field $v_0 \in V$, the RKHS associated to K_G , and for any $\tau \in [0,1]$, there exist a time parametrization γ_τ and a distortion h_τ such that:

$$\exp_{Id}(\tau v_0) \cdot G(s_0) = G((s_0 + h_\tau) \circ \gamma_\tau^{-1}).$$

Experiments

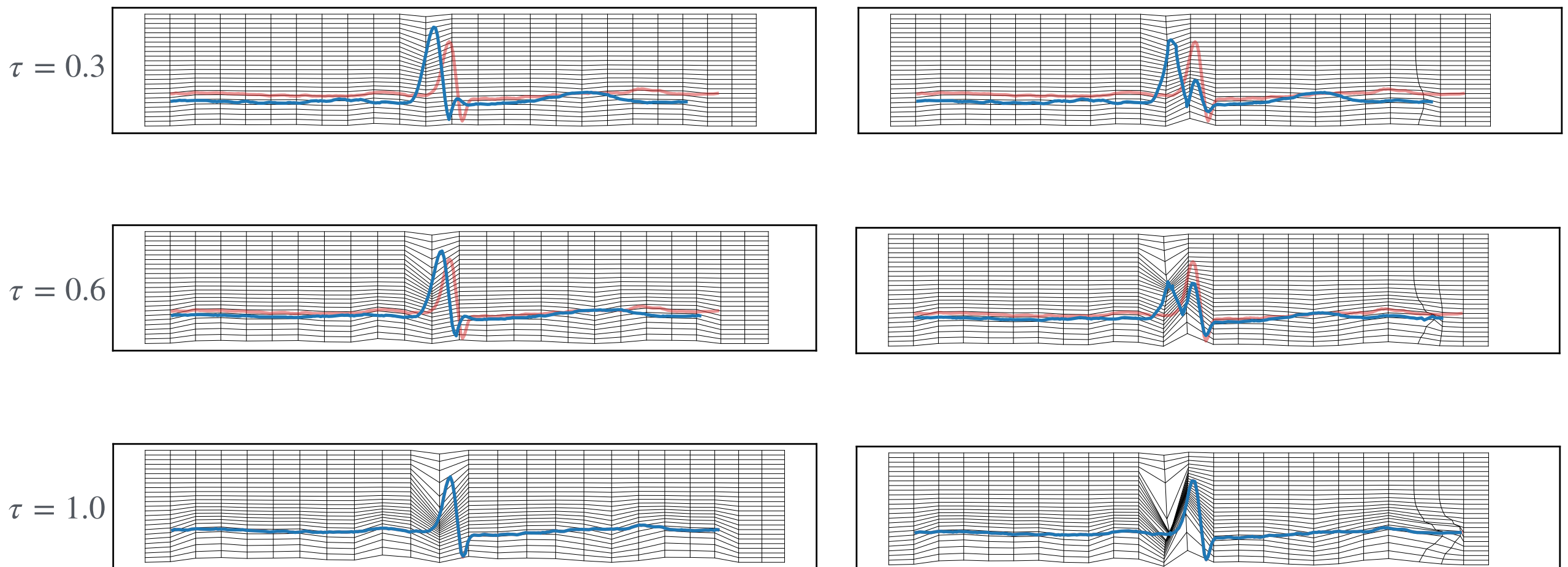
Example of transport

ECG heartbeat



TS-LDDMM (ours)

LDDMM



Robustness to irregular sampling

Experimental protocol:

- Task: Classification with different scenarios of irregular sampling.
- Datasets: 15 UCR/UEA datasets.
- Metric: f1-score.
- Algorithms: TS-LDDMM, LDDMM and 8 deep learning algorithms.

Results:

First & second best performers. TS-LDDMM is the best performer on three out of four dropping rate.

Methods	Regular		30 % dropped		50 % dropped		70 % dropped	
	F1-score	Rank	F1-score	Rank	F1-score	Rank	F1-score	Rank
RNN (1999)	0.64 ± 0.21	6.2	0.53 ± 0.23	6.6	0.48 ± 0.21	7.2	0.44 ± 0.21	6.07
LSTM (1997)	0.61 ± 0.29	6.0	0.57 ± 0.29	6.27	0.53 ± 0.25	6.07	0.51 ± 0.29	5.27
GRU (2014)	0.71 ± 0.26	4.2	0.68 ± 0.28	4.27	0.66 ± 0.28	3.73	<u>0.59 ± 0.28</u>	<u>3.67</u>
MTAN (2021)	0.59 ± 0.28	7.13	0.58 ± 0.28	5.8	0.54 ± 0.29	5.33	0.51 ± 0.28	5.0
MIAM (2022)	0.48 ± 0.35	6.93	0.42 ± 0.33	8.27	0.47 ± 0.31	6.93	0.35 ± 0.31	7.6
ODE-LSTM (2020)	0.63 ± 0.24	6.0	0.57 ± 0.25	6.53	0.51 ± 0.24	7.27	0.45 ± 0.23	6.73
Neural SDE (2019)	0.48 ± 0.28	7.67	0.47 ± 0.26	7.47	0.45 ± 0.27	7.13	0.45 ± 0.25	6.0
Neural LNSDE (2024)	0.7 ± 0.27	<u>3.87</u>	0.68 ± 0.29	<u>4.0</u>	<u>0.67 ± 0.25</u>	<u>3.53</u>	0.66 ± 0.23	2.47
LDDMM (2008)	<u>0.72 ± 0.2</u>	4.53	<u>0.7 ± 0.21</u>	4.2	0.57 ± 0.25	5.0	0.4 ± 0.25	7.13
TS-LDDMM (ours)	0.83 ± 0.18	2.93	0.8 ± 0.18	2.07	0.7 ± 0.26	3.33	0.51 ± 0.27	5.67

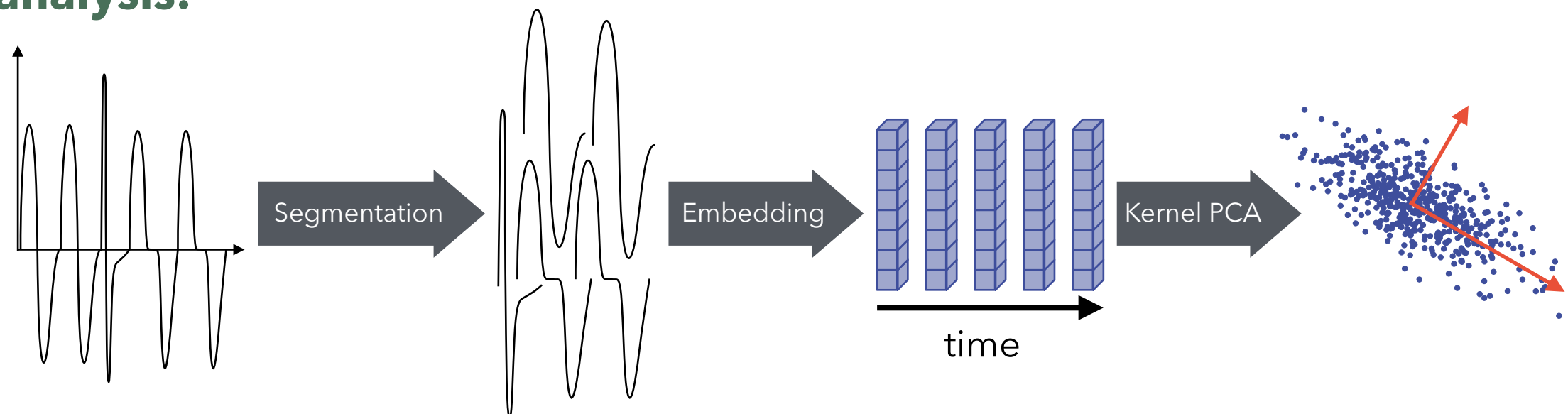
Mouse ventilation analysis

Aim: Assessing ventilation response after the exposure of a neurotoxin for mice of different genotypes (WT and ColQ).

Experimental protocol: For each genotype (WT, ColQ):

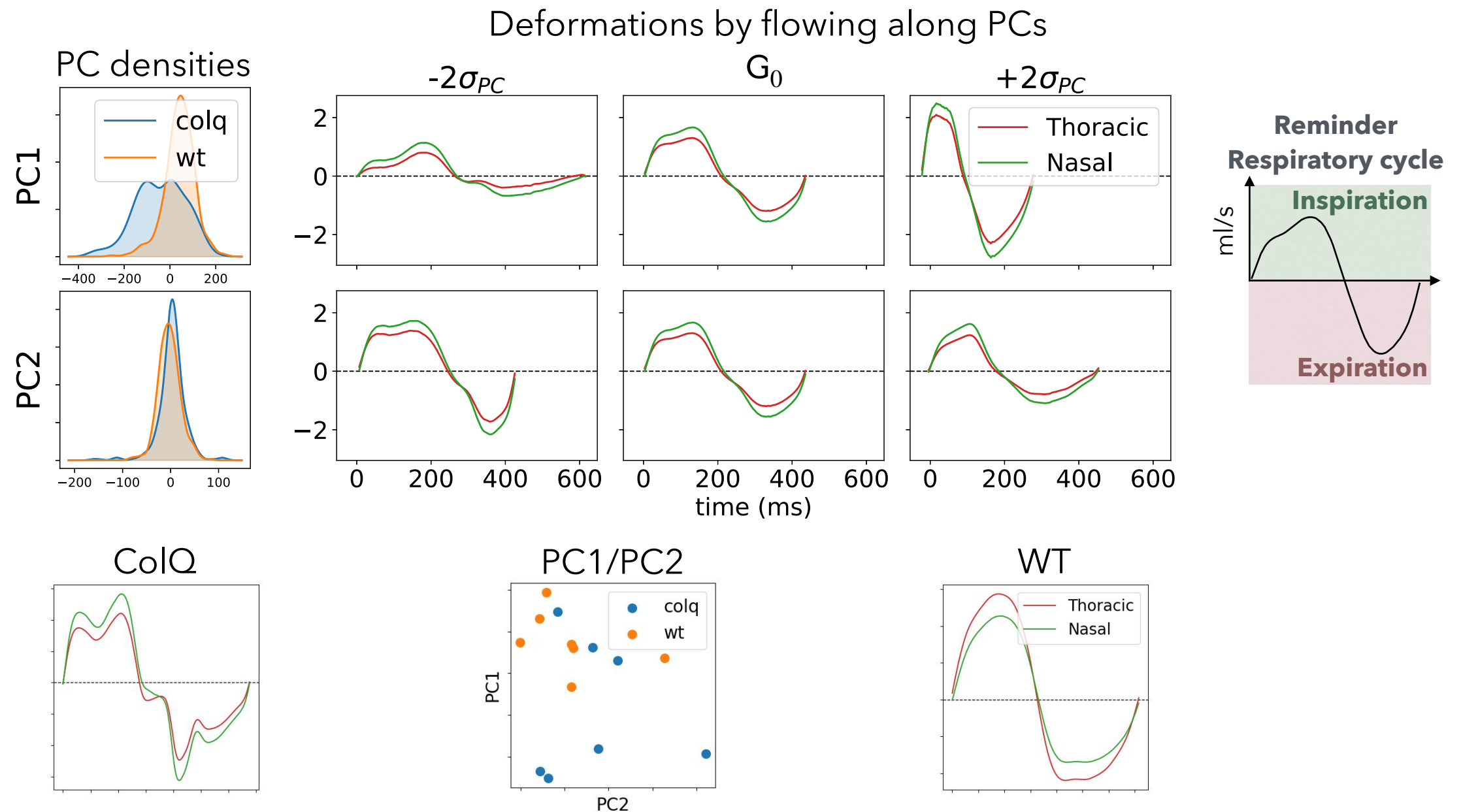


Data analysis:



Control period analysis

Kernel PCA applied on the initial velocity fields parametrized with $(\mathbf{G}_0, \alpha_j)_{j \in [N]}$.

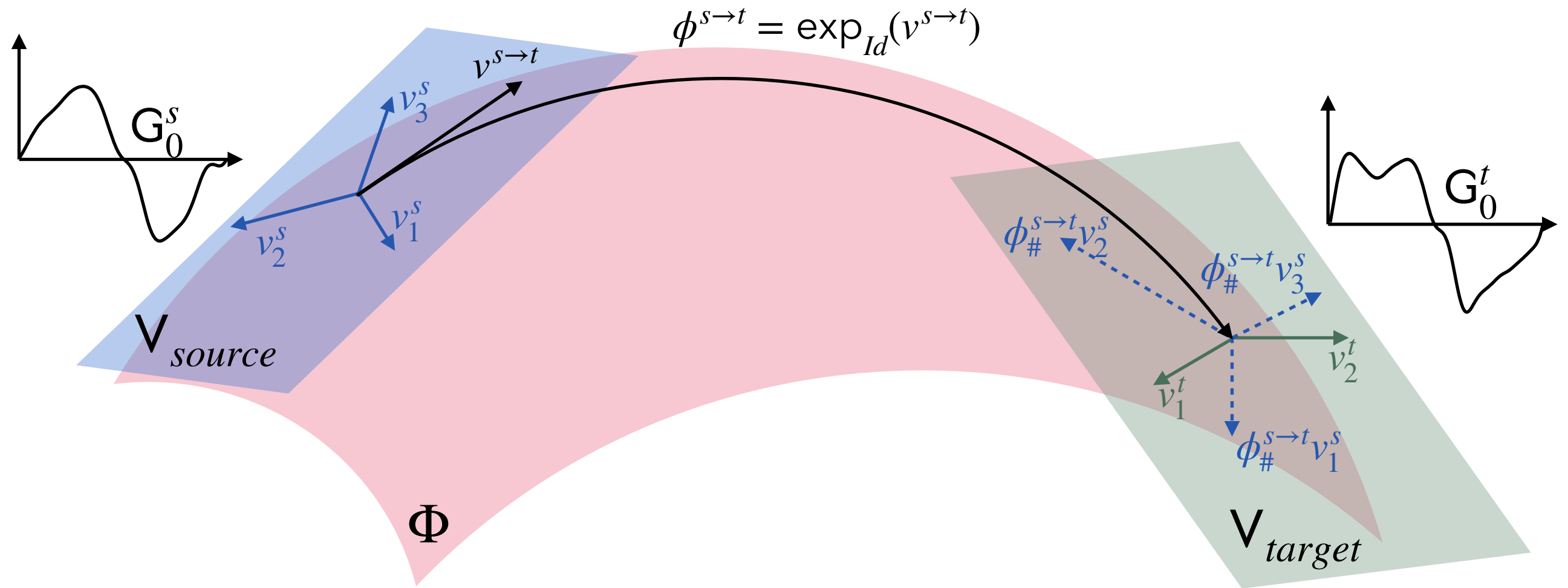


Observations:

- PC1 accounts for the cycle duration and PC2 for the ratio inspiration/expiration.
- PC1 negative tail for ColQ mice seems to highlight impairment of motor control.

Going further

Domain Adaptation



Parallel transport: a tool to transport vectors from one tangent space to another. Exist on the space of diffeomorphisms Φ [BE05].

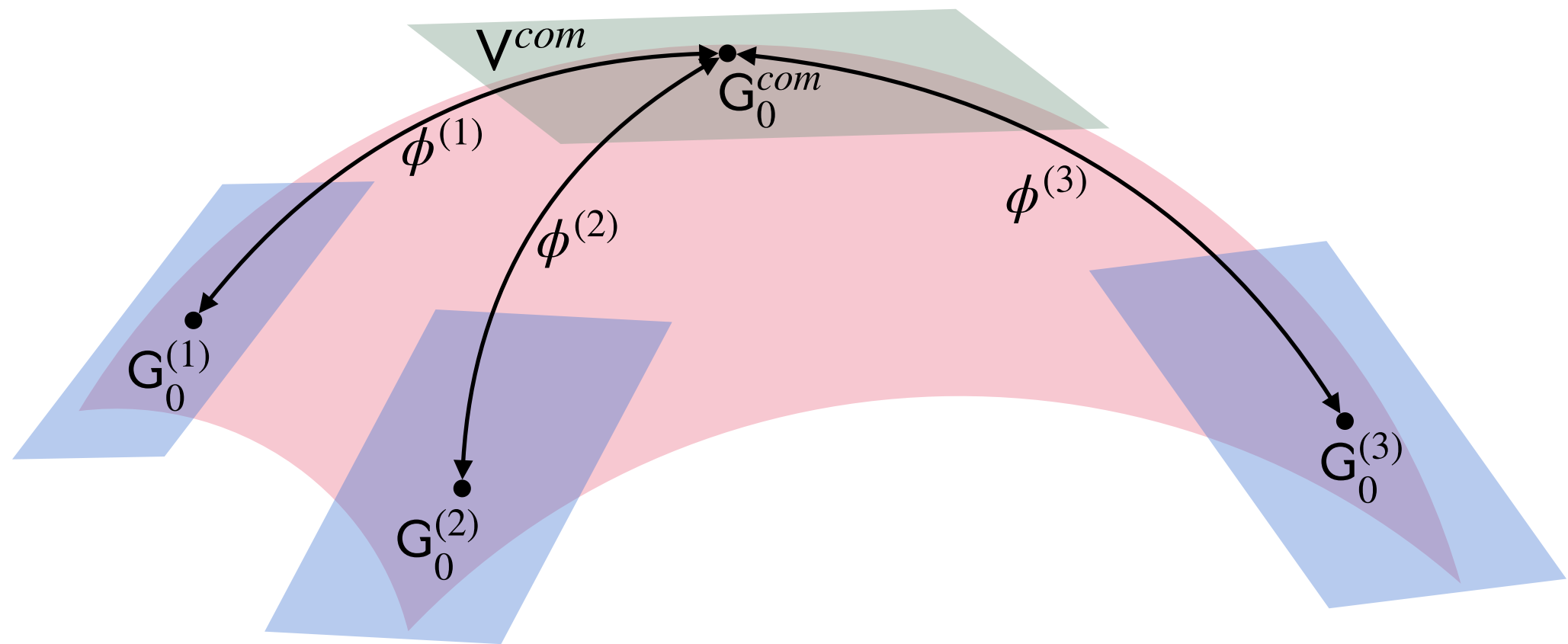
Suggestion:

Input: source and target reference graphs, a predictor learned on source vectors.

1. Perform the registration between the source and target reference graphs: $\phi^{s \rightarrow t}$
2. Parallel transport source vectors

Output: source predictor adapted to target domain

Personalized Model Learning



Following the idea of parallel transport and domain adaptation.

Suggestion:

Input: reference graphs $\{G_0^{(1)}, \dots, G_0^{(N)}\}$ and respective tangent vectors.

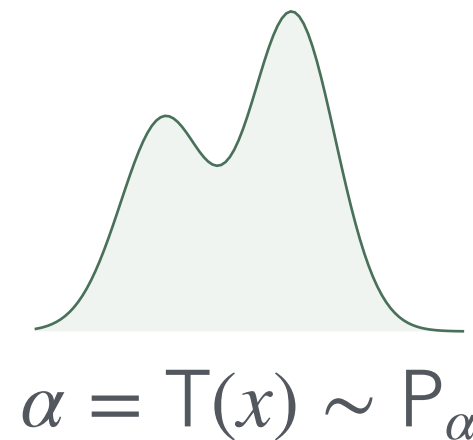
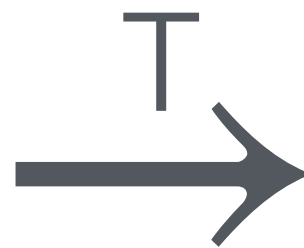
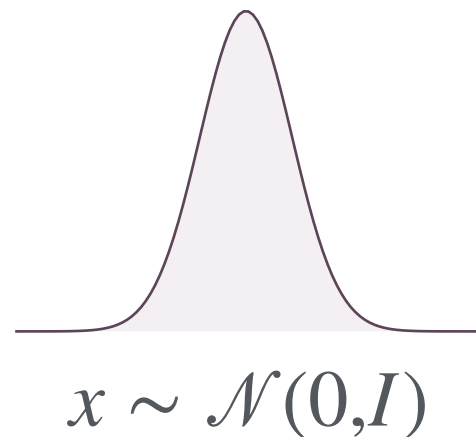
1. Estimate the common reference graph G_0^{com} from $\{G_0^{(1)}, \dots, G_0^{(N)}\}$.
2. Parallel transport vectors from all tangent spaces to G_0^{com} .
3. Learn a common predictor.
4. Push the common model on individual domains by domain adaptation.

Output: Domain specific predictors

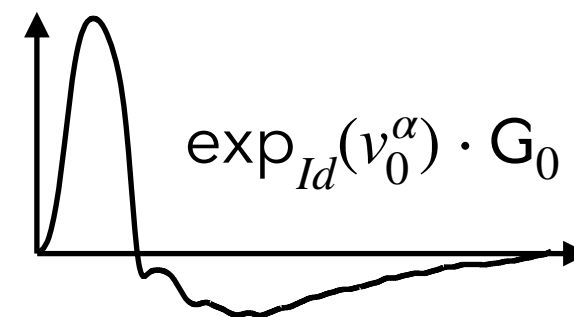
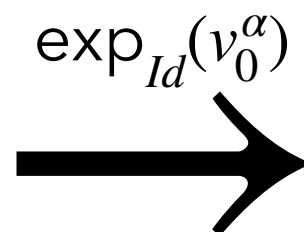
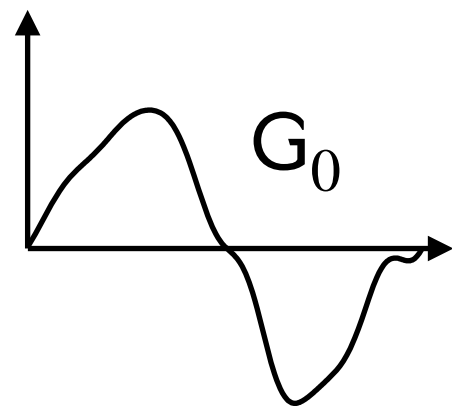
Generative Modeling

Recall: the embedding learning outputs a graph $G_0 \in \mathbb{R}^{m \times (d+1)}$, and momentums $(\alpha_k^0)_{k \in [N]} \in (\mathbb{R}^{m \times (d+1)})^N$.

Let:



Then:



Suggestion: Suppose that $\alpha \sim P_\alpha$.

1. Learn T such that $T_\# \mathcal{N}(0, I) = P_\alpha$.
2. Sample x from $\mathcal{N}(0, I)$.
3. Set $\alpha = T(x)$.
4. Perform a geodesic shooting from G_0 with initial velocity field v_0^α .

Operator-based embeddings

How can we compare operator embeddings of dynamical systems ?

A Spectral-Grassmann Wasserstein metric for operator representations of dynamical systems

Germain, T., Flamary, R., Kostic, V. R., & Lounici, K.
arXiv preprint, 2025.

Motivation

Dynamical system & machine learning

Dynamical systems are backbone models of temporally evolving phenomena:

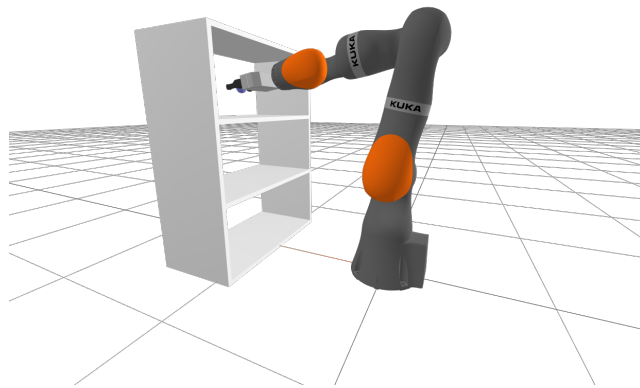
Modeling:

$$x_{t+1} = g(t, x_t)$$

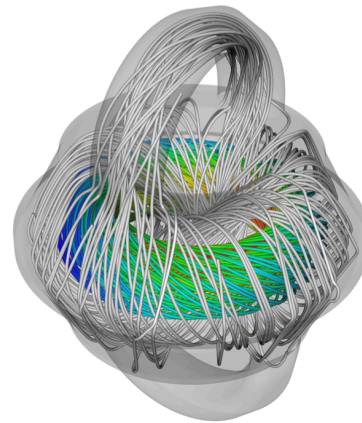
discrete time

$$\frac{dx(t)}{dt} = g(t, x(t))$$

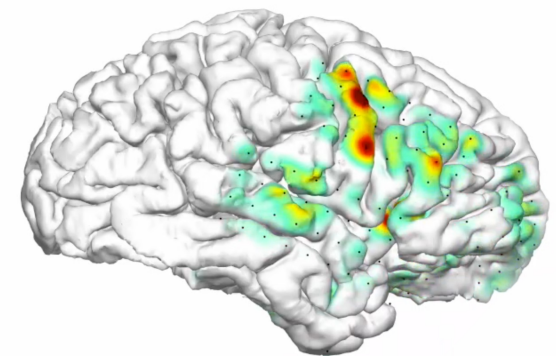
continuous time



Robotic



Fluid dynamic



Brain dynamic

Paradigm shift:

- Classical approach: ODE/PDE/SDE design + parameter fitting
- ML approach : Can we build dynamical models purely from observed data ?

➡Elegant formulation through transfer operator theory.

Learning & Interpreting dynamical systems

Transfert operator theory

Definition (transfer operator). Let assume:

- A stochastic process: $(X_t)_{t \geq 0} \subseteq \mathcal{X}$
- A real-valued functional space: $\mathcal{F} \subset \mathbb{R}^{\mathcal{X}}$

Under some additional assumptions, for any $t \geq 0$, there exists a linear operator $A_t : \mathcal{F} \rightarrow \mathcal{F}$, called *transfer operator*, such that:

$$[A_t(f)](x) \triangleq \mathbb{E}[f(X_t) | X_0 = x], \quad x \in \mathcal{X} \text{ and } f \in \mathcal{F}.$$

Remark: Even if the dynamical system is non-linear, the transfert operator is linear.

Time-homogeneous systems: Since $A_{t+s} = A_t A_s$:

(discrete): $A_t = A_1^t, \quad \forall t \in \mathbb{N}$

(continuous): $A_t = \exp(tL), \quad \text{with} \quad L = \lim_{t \rightarrow 0} (A_t - Id)/t \quad (\text{infinitesimal generator})$

➡ It suffices to estimate $A = A_1$ or L .

Spectral decomposition

Spectral decomposition: Assuming that $\mathcal{F}$ is a separable Hilbert space (typically $L^2_\pi(\mathcal{X})$ with π the system's invariante measure) and L is a non defective operator with purely discrete spectrum. Then L can be written as:

$$L = \sum_{j \in \mathbb{N}} \lambda_j g_j \otimes f_j, \quad \text{s.t.} \quad Lf_j = \lambda_j f_j, \quad L^* g_j = \bar{\lambda}_j g_j, \quad \langle f_i, g_j \rangle = \delta_{i,j}$$

Intuition: Suppose $\text{rank}(L) = r$ and $\mathcal{F}$ is d -dimensional with $r < d$, then:

$$L = \underbrace{\begin{pmatrix} | & & | \\ f_1 & \cdots & f_r \\ | & & | \end{pmatrix}}_F \underbrace{\begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_r \end{pmatrix}}_\Lambda \underbrace{\begin{pmatrix} - & g_1 & - \\ & \vdots & \\ - & g_r & - \end{pmatrix}}_{H^*} \quad \text{with} \quad H^* F = I_r$$

Benefit:

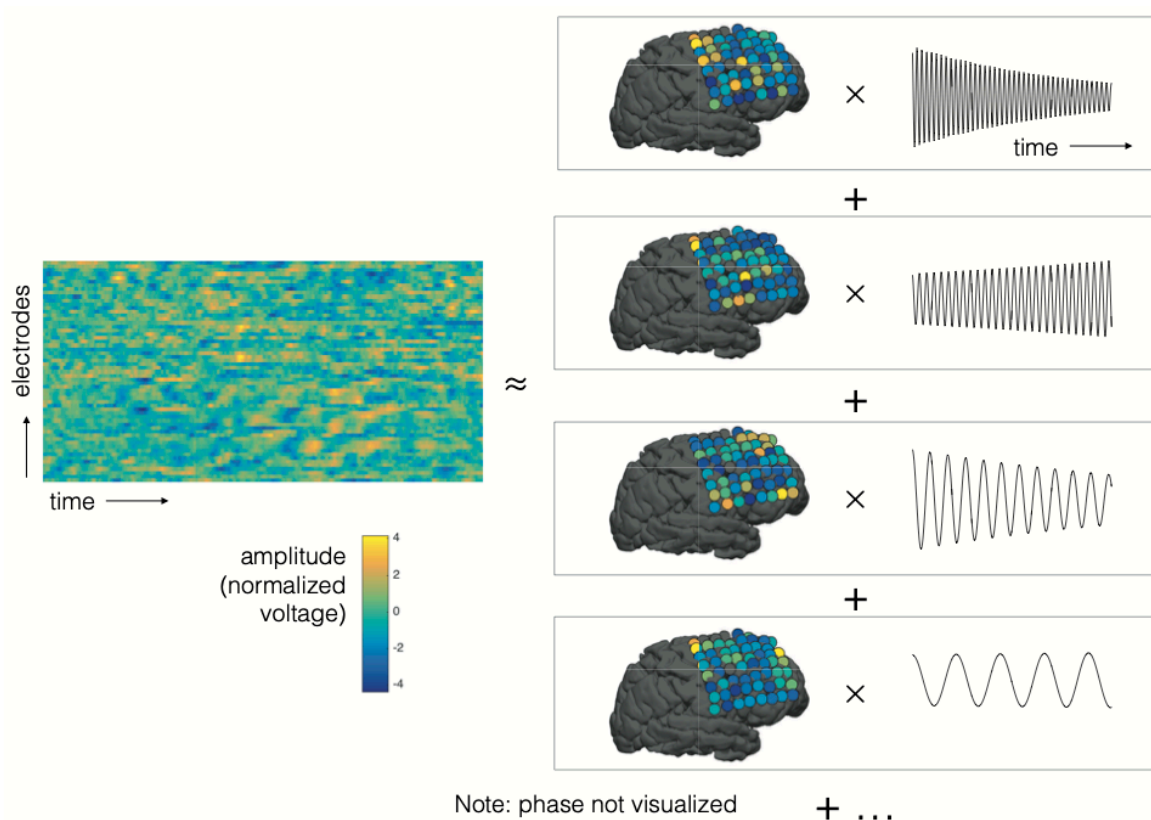
- Fast computation: $[A_t f](x) = \mathbb{E}[f(X_t) | X_0 = x] = [F \exp(t\Lambda) H^* f](x)$
- Evolves densities: if $X_0 \sim q_0$ then $X_t \sim A_t^* q_0 = H \exp(t\Lambda) F^* q_0$

Spectral decomposition: Interpretation

Space and time disentangled: The spectral decomposition of $A_t = \exp(tL)$ decouples the evolution of any arbitrary observable $f \in \mathcal{F}$:

$$[A_t f](x) = \mathbb{E}[f(X_t) | X_0 = x] = \sum_{j \in \mathbb{N}} e^{\lambda_j t} \langle g_j, f \rangle f_j(x) = \sum_{j \in \mathbb{N}} e^{\tau_j t + i 2\pi \omega_j t} m_j^f(x),$$

into **temporal** and **static** components.

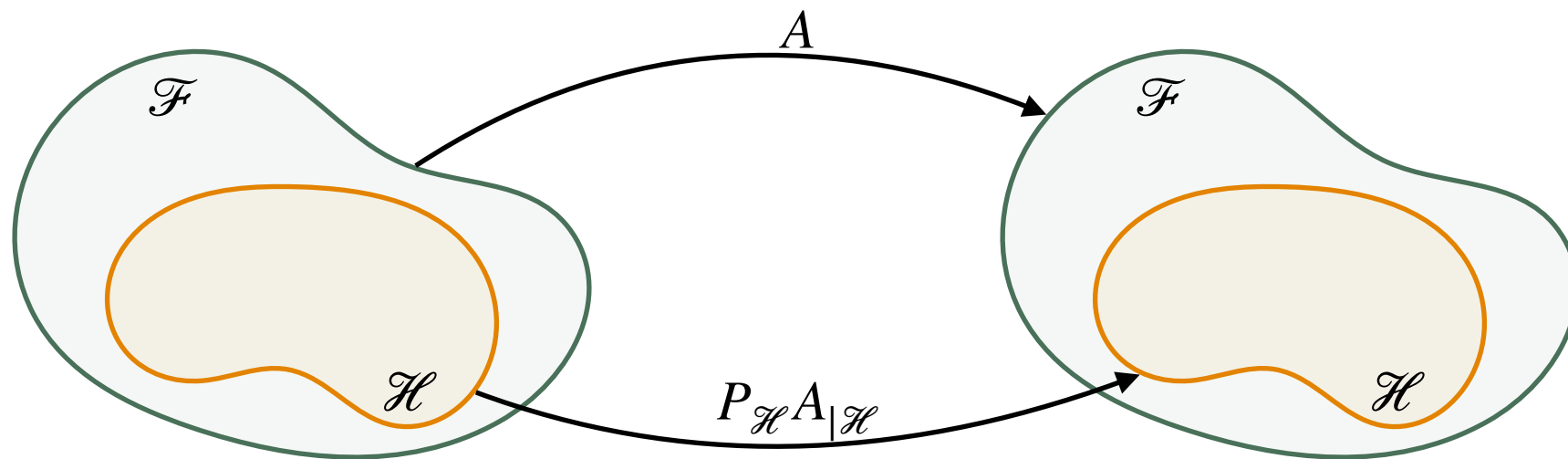


- $\tau_j = \mathcal{R}(\lambda_j)$ are dynamics' timescales.
- $\omega_j = \mathcal{I}(\lambda_j)/2\pi$ are oscillating frequencies.
- $m_j^f = \langle g_j, f \rangle f_j$ are **Koopman modes** of f .

Brain dynamic mode decomposition [Kut16]

Learning transfer operators

Problem: Only trajectories of dynamical systems are observed. Neither the operator A nor its domain $\mathcal{F}$ are known.



Popular approach [Bru22, Kos22]: Learning the action of $A : \mathcal{F} \rightarrow \mathcal{F}$ on a predefined RKHS $\mathcal{H} \subset \mathcal{F}$ with kernel $\kappa(x, y) \triangleq \langle \phi(x), \phi(y) \rangle$ via empirical risk minimization of the projected operator $P_{\mathcal{H}} A|_{\mathcal{H}}$, i.e.:

$$\operatorname{argmin}_{G \in HS(\mathcal{X})} \frac{1}{n} \sum_{i \in [n]} \|\phi(y_i) - G^* \phi(x_i)\|_{\mathcal{H}}^2$$

with the observation dataset $\{(x_i, y_i)\}_{i \in [n]}$ (typically a trajectory $y_i = x_{i+1}$).

Remark: Spectral estimation guarantees for universal bounded kernels [Kos23].

Open problem

Observation:

While the estimating individual dynamical systems has been thoroughly explored, theoretically grounded and computationally efficient **methods** for **comparing** different dynamical **systems** remain **under explored**.

Related work:

- Martin distance [Mar02] : pseudo-metric on ARMA models
- Binet-Cauchy kernel [Vid13]: Martin distance extension to LDS
- Topological conjugacy [Red24]: Optimal Transport on spectrum
- Hilbert-Schmidt and operator norms are too conservative

➡ Introducing Spectral-Grassmann Wasserstein Metric (SGOT)

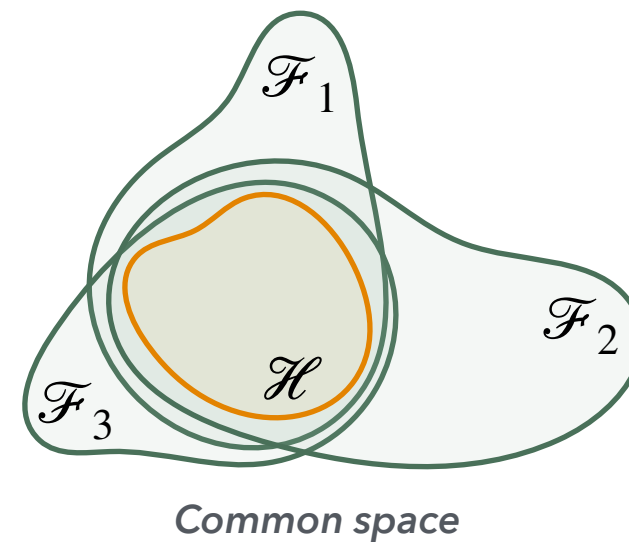
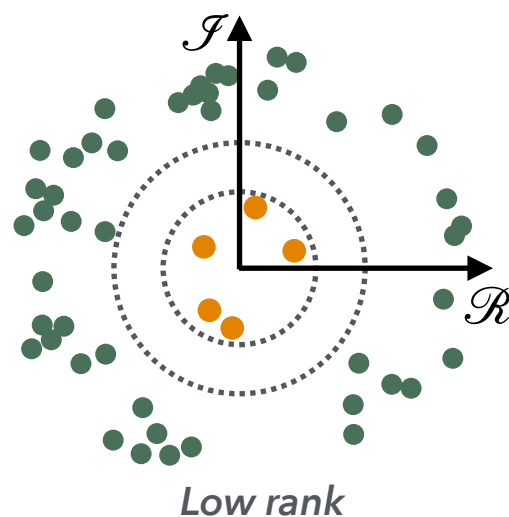
Spectral-Grassmann Wasserstein Metric (SGOT)

SGOT intuition & assumptions

Intuition: Back to spectral decomposition:

$$L = \sum_{i \in \mathbb{N}} \sum_{j \in [n_i]} \lambda_i g_{i,j} \otimes f_{i,j} = \sum_{i \in \mathbb{N}} \lambda_i P_i, \quad \text{s.t.} \quad \langle f_{i,j}, g_{u,v} \rangle = \delta_{(i,j),(u,v)}$$

➔ Spectral decomposition are **unique up to a permutation** (optimal transport) and a **change of basis** (Grassmann manifold) for spectral projectors (P_i).

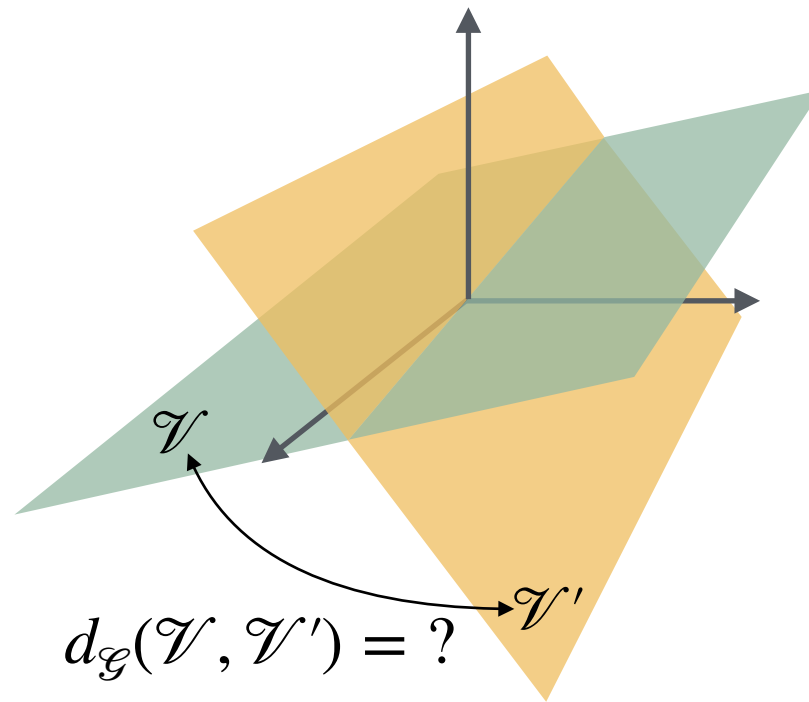


Assumptions:

1. **(Time homogeneous Markovian dynamical systems):** Let consider $\{L_k\}_{k \in [N]}$.
2. **(Low rank operators):** For any $k \in [N]$ there exists $r_k \in \mathbb{N}$ such that the r_k eigenvalues the closest to the origin are separated from all the others.
3. **(A common functional space):** There exists an RKHS $\mathcal{H}$ such that for any operator, the estimation of its r_k -restriction, $T_k = \exp(L_k|_{r_k})$, is well defined.

Grassmann Manifold

Aims: Comparing vector subspaces independently of their representative basis.

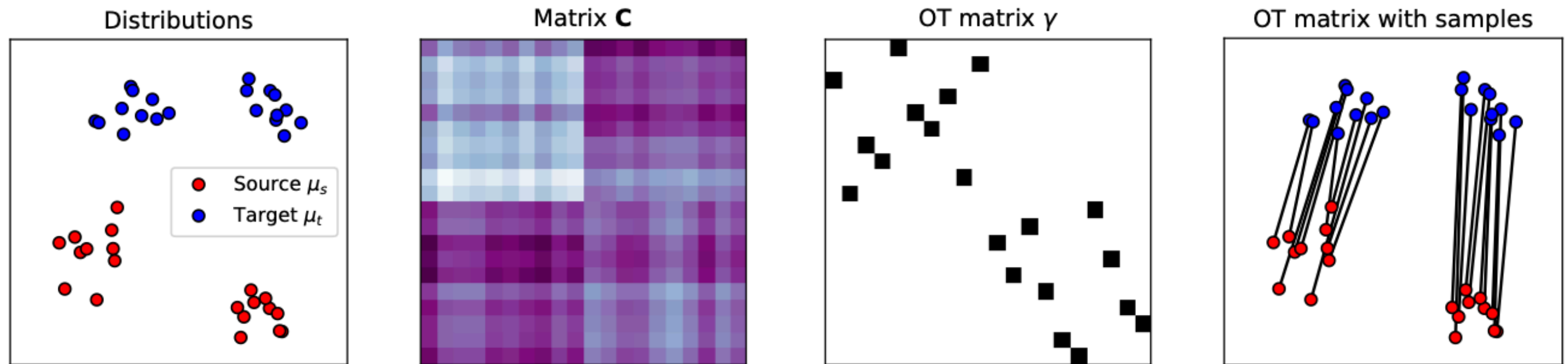


Proposition (Extension to infinite dimension space): Let $\mathcal{H}$ be a separable Hilbert space, and $\mathcal{G}_r(\mathcal{H})$ the set of vector subspaces of with dimension at most r . With the application:

$$d_{\mathcal{G}}(\mathcal{V}, \mathcal{V}') = \|P_{\mathcal{V}} - P_{\mathcal{V}'}\|_{\mathcal{H}} \quad \text{for any } (\mathcal{V}, \mathcal{V}') \in \mathcal{G}(\mathcal{H})^2,$$

where P_* are orthogonal projectors, $\mathcal{G}_r(\mathcal{H})$ is a separable complete metric space.

Discrete Optimal Transport



Kantorovich formulation:

Given two probability distributions $\mu_s = \sum_{i \in [n_s]} a_i \delta_{x_i^s}$ and $\mu_t = \sum_{i \in [n_t]} b_i \delta_{x_i^t}$, the optimal transport cost of order $p \in \mathbb{N}^*$ verifies:

$$W_{d,p}^p(\mu_s, \mu_t) = \min_{P \in \Pi(\mu_s, \mu_t)} \langle C, P \rangle_F,$$

where C is the cost matrix $c_{ij} = d(x_i^s, x_j^t)^p$, and the transport matrix P is from the set:

$$\Pi(\mu_s, \mu_t) = \{P \in \mathbb{R}_+^{n_s \times n_t} \mid P1_{n_t} = a, P^\top 1_{n_s} = b\}$$

Remarks:

- Computable with network simplex in $n^3 \log(n)$ where $n = n_s = n_t$.
- $W_d^p(\mu_s, \mu_t)$ is called a Wasserstein metric.

Transfer operator as probability distribution

Recall: Let $\mathcal{H}$ be a separable $\mathbb{C}$ -Hilbert space and $\mathcal{S}_r(\mathcal{H})$ the set of non-defective operator on $\mathcal{H}$ with rank at most $r \in \mathbb{N}$. For any $L \in \mathcal{S}_r(\mathcal{H})$, its spectral decomposition can be written as:

$$L = \sum_{i \in [\ell]} \sum_{j \in [m_i]} \lambda_i g_{i,j} \otimes f_{i,j} \quad \text{s.t.} \quad \langle f_{i,j}, g_{u,v} \rangle = \delta_{(i,j),(u,v)} \quad \& \quad m_{tot} = \sum_{i \in [\ell]} m_i \leq r .$$

Property (distribution embedding): Transfer operator embedding as probability distribution: The application defined on $\mathcal{S}_r(\mathcal{H})$ by:

$$\mu(L) \triangleq \sum_{i \in [\ell]} \frac{m_i}{m_{tot}} \delta_{(\lambda_i, \mathcal{V}_i)} ,$$

where $\mathcal{V}_i = \text{span}(\{(g_{i,j} \otimes f_{i,j}) \mid j \in [m_i]\})$, is an injective embedding.

Theorem (Wasserstein metric): Let $\eta \in (0,1)$, and consider the elementary metric:

$$d_\eta((\lambda, \mathcal{V}), (\lambda', \mathcal{V}')) \triangleq \eta |\lambda - \lambda'| + (1 - \eta) d_{\mathcal{G}}(\mathcal{V}, \mathcal{V}') .$$

Then $\mathcal{S}_r(\mathcal{H})$ endowed with the application:

$$d_{\mathcal{S}}(L, L') \triangleq W_{d_\eta, p}(\mu(L), \mu(L')) ,$$

defines a metric space.

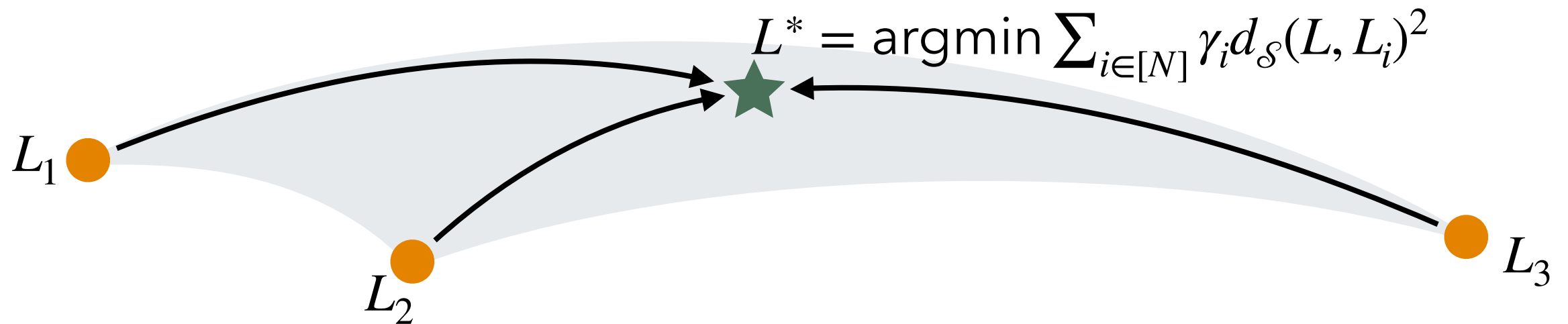
Metric estimator's converge bound

Recall: Under kernel universality assumption such methods come with spectral estimation guarantees [Kos23], specifically for the Reduced Rank Regression (RRR) method [Kos22].

Theorem (simplified): Suppose two dynamical systems with low rank operator projection $L_1, L_2 \in \mathcal{S}_r(\mathcal{H})$ and $\hat{L}_1, \hat{L}_2$ their estimation from n samples with the RRR method [Kos22]. Suppose $\alpha \in (1, 2)$, and $\beta \in [0, 1]$ bounding the empirical covariance. In the i.i.d setting, for any $\delta \in (0, 1)$, with probability $1 - \delta$ it holds:

$$|d_{\mathcal{S}}(\hat{L}_1, \hat{L}_2) - d_{\mathcal{S}}(L_1, L_2)| \leq n^{-\frac{\alpha-1}{2(\alpha+\beta)}} \ln(2\delta^{-1})$$

SGOT Barycenter



Fréchet mean problem: Given $\{L_i\}_{i \in [N]}$ and $\gamma \in \mathbb{R}_+^N$ s.t. $\sum_{i \in [N]} \gamma_i = 1$, we aim to solve:

$$\operatorname{argmin}_{L \in \mathcal{S}_r(\mathcal{H})} \sum_{i \in [N]} \gamma_i d_{\mathcal{S}}(L, L_i)^2$$

Constrained optimization problem: Considering the parametric operator:

$$L_{\theta} \triangleq \sum_{i \in [r]} \lambda_i \kappa_x \beta_i \otimes \kappa_x \alpha_i$$

with $\lambda \in \mathbb{C}^r$, $x \in \mathcal{X}^n$ and $\alpha, \beta \in \mathbb{C}^{n \times r}$, the problem becomes:

$$\operatorname{argmin}_{P, \theta} \sum_{i \in [N]} \gamma_i \langle C_i(\theta), P_i \rangle \quad \text{s.t.} \quad \begin{cases} K = \{\kappa(x_i, x_j)\}_{(i,j) \in [n]^2} \\ \alpha^* K \beta = I \\ P_i \in \Pi(\mu(L_{\theta}), \mu(L_i)), \forall i \in [N] \end{cases}$$

➔ *Optimization: Riemannian gradient descent or block coordinates descent.*

Experiments

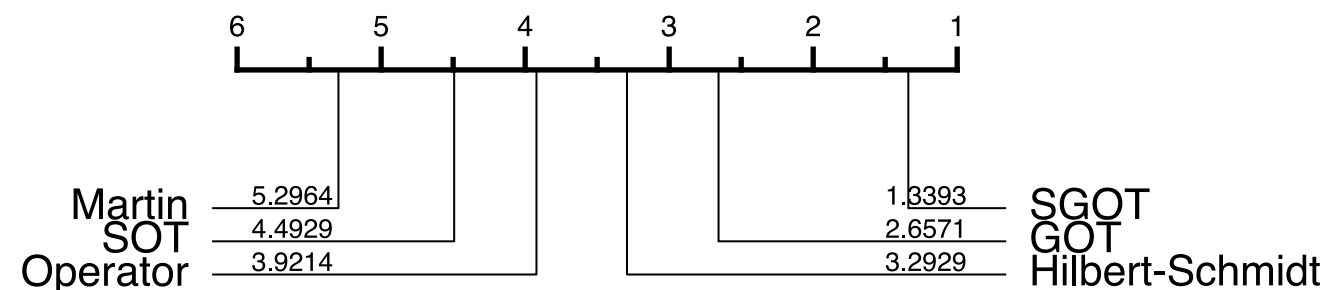
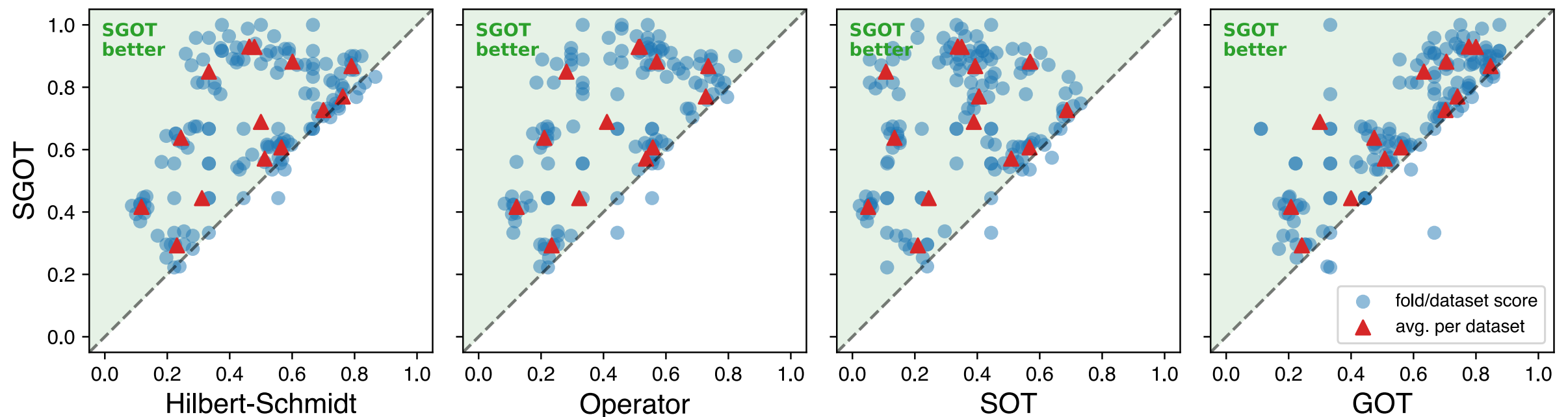
Classification

Dataset: 14 datasets from the UEA time series database.

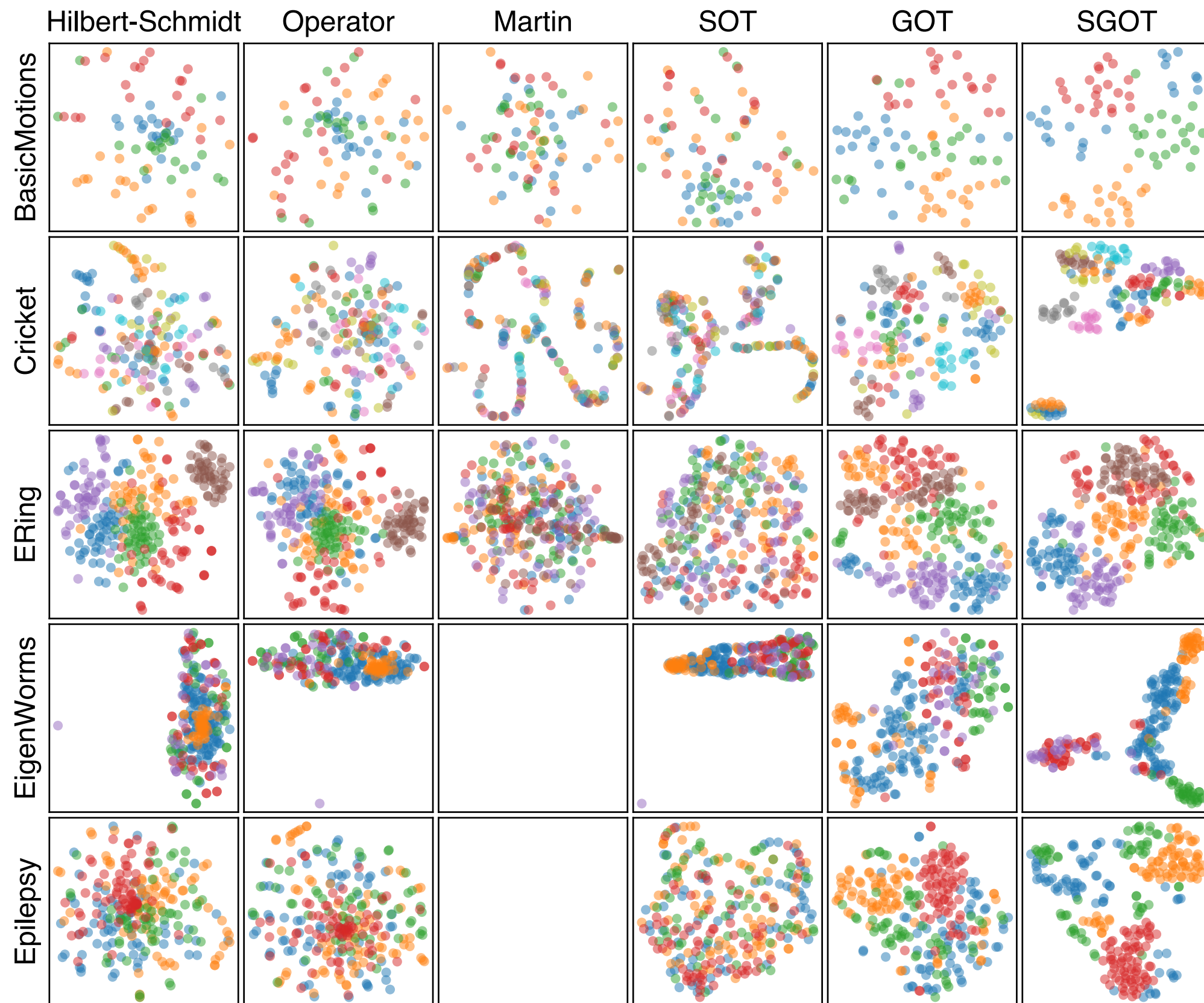
Classifier: K-nearest-neighbors with 6 different metrics.

Score function: Accuracy.

Protocol: Nested cross-validation + grid search for neighbors K and SGOT ratio η .

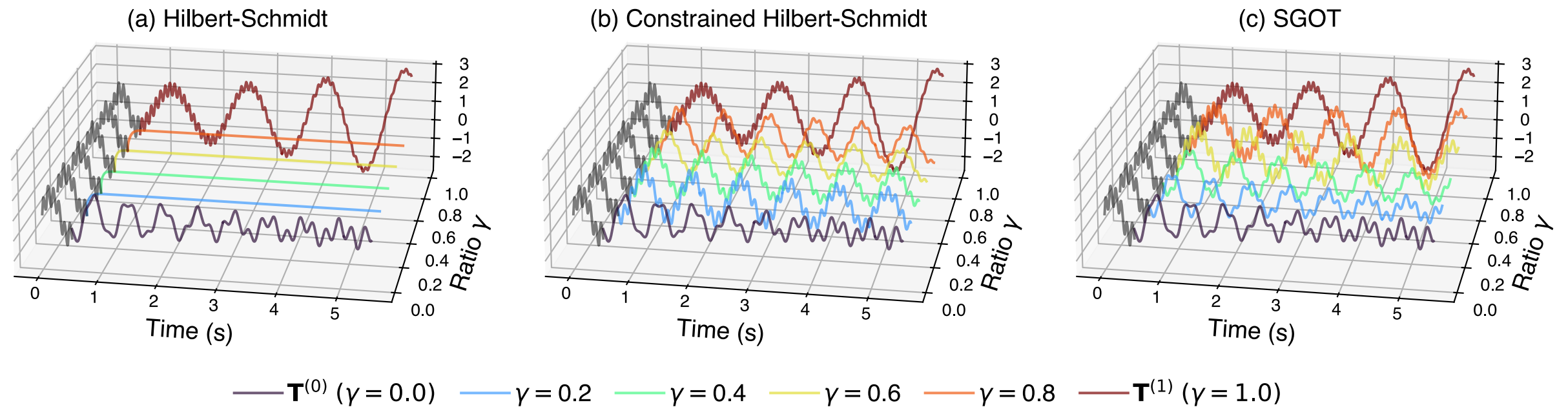


T-SNE embeddings

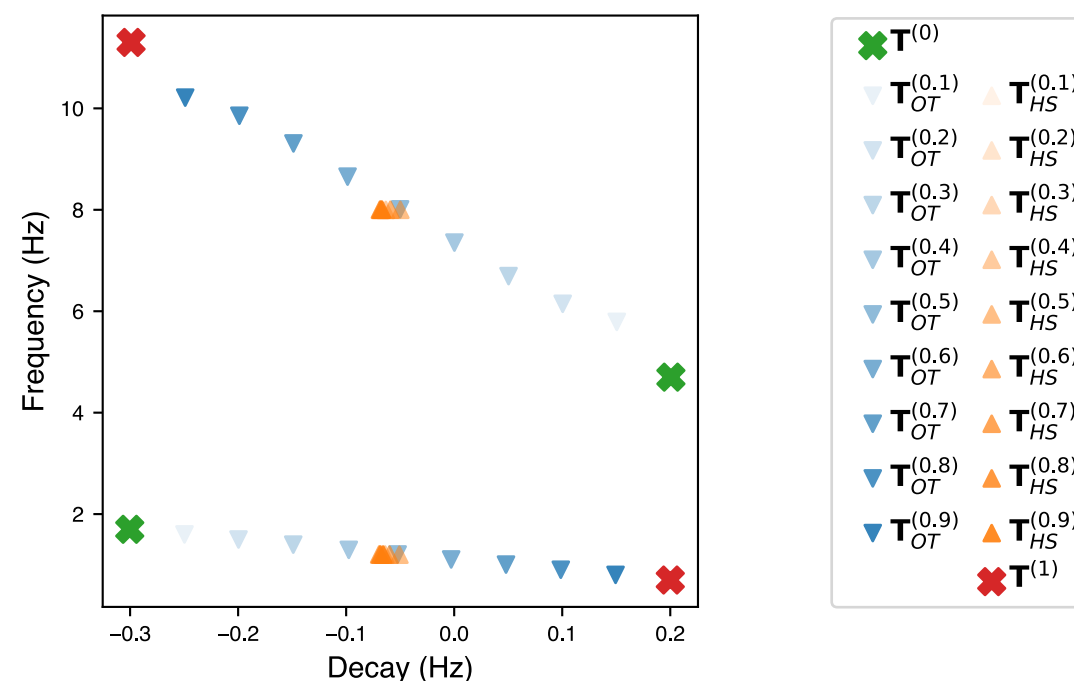


Barycenter & Interpolation

Barycentric interpolation between two linear dynamical systems.



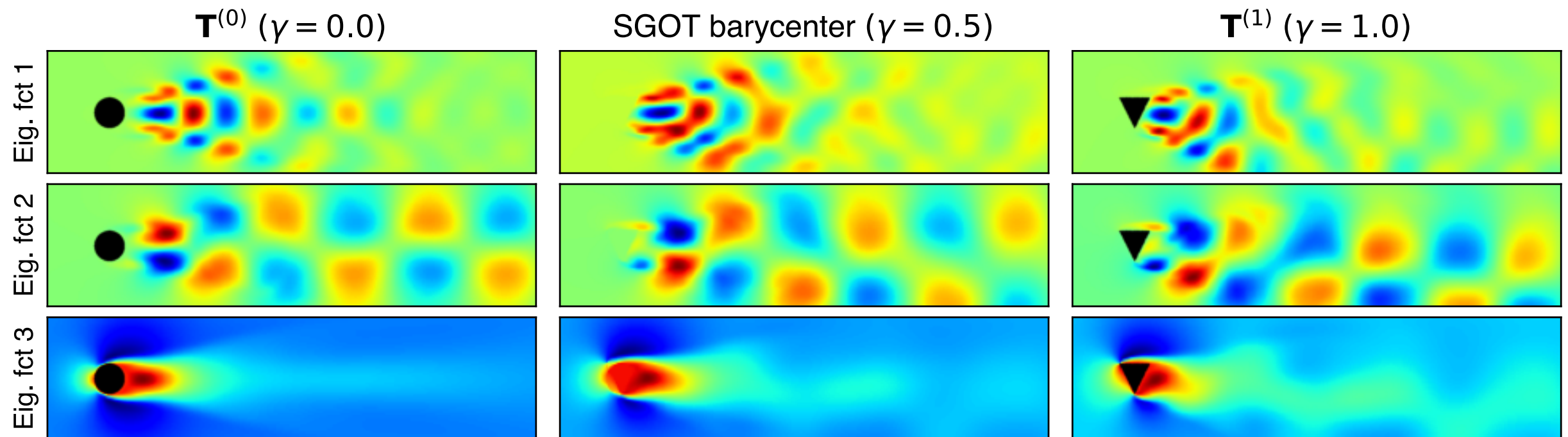
Barycenters' eigenvalues evolution:



Barycenter & Interpolation

Barycentric interpolation between two fluid dynamics.

Fluid flow from left to right through an obstacle (circle or triangle)



Going Further

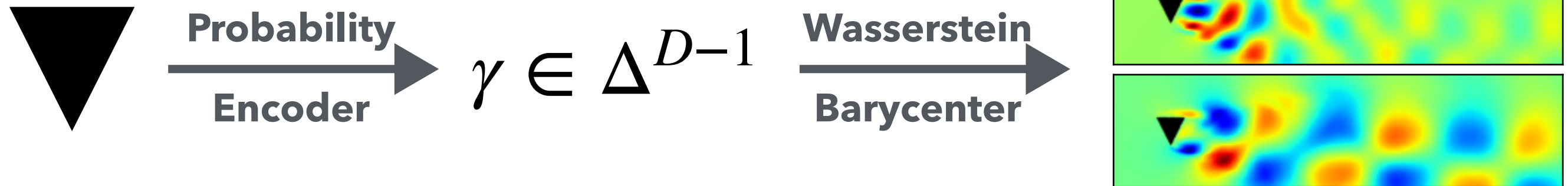
Wasserstein dictionary learning

Aim: From a set $\{L_i\}_{i \in [N]}$ learn a dictionary of transfer operators $\{\bar{L}_i\}_{i \in [D]}$ such that for any L_i there exists $\gamma \in \Delta^{D-1}$ verifying:

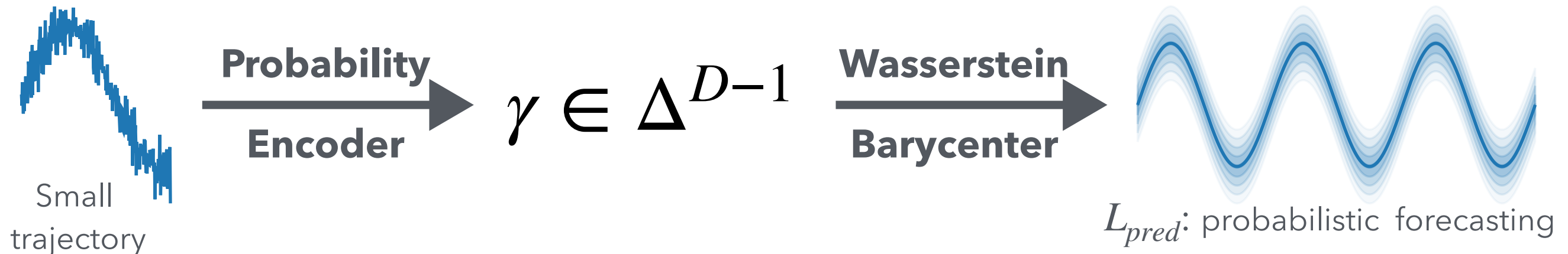
$$L_i \approx \operatorname{argmin}_{L \in \mathcal{S}_r(\mathcal{H})} \sum_{j \in [D]} \gamma_j d_{\mathcal{S}}(L, \bar{L}_j)^2$$

Downstream application:

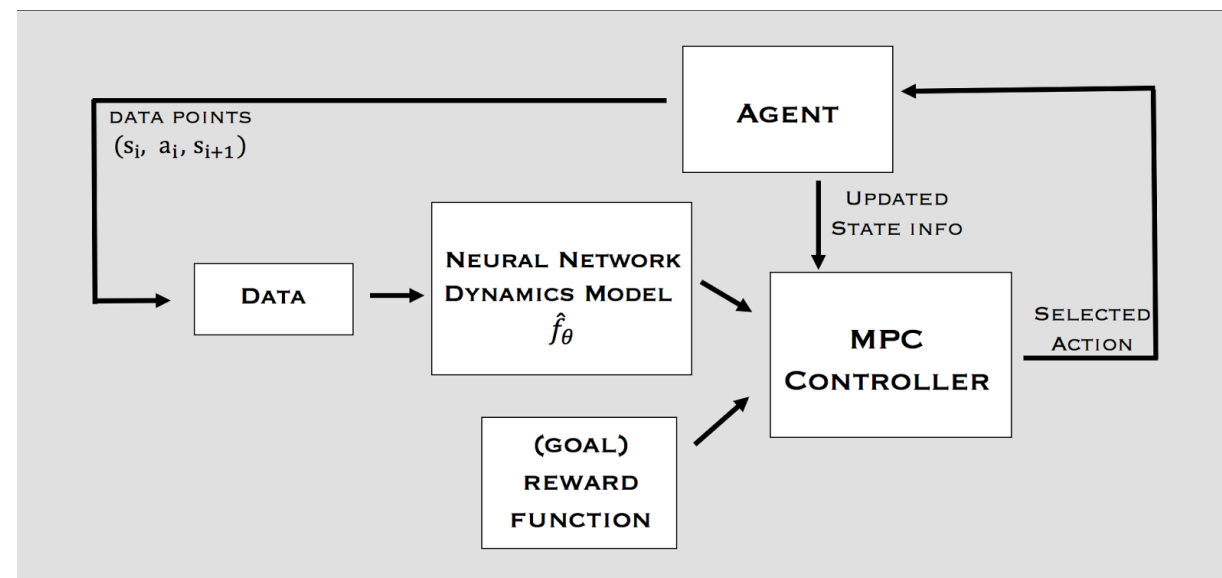
- Physical systems predictions:



- Early system prediction:



Model-based reinforcement learning



Model-based RL framework, from [AGR17]

Current state: Control approach exist, leveraging the transfert operator of the control flow, with the control variable $\mathbf{u} : [0,1] \mapsto \mathcal{C}$:

$$[A_t^{\mathbf{u}}(f)](x) \triangleq \mathbb{E}[f(X_t, u_t) | X_0 = x, U = \mathbf{u}], x \in \mathcal{X} \text{ and } f \in \mathcal{F}$$

Open questions:

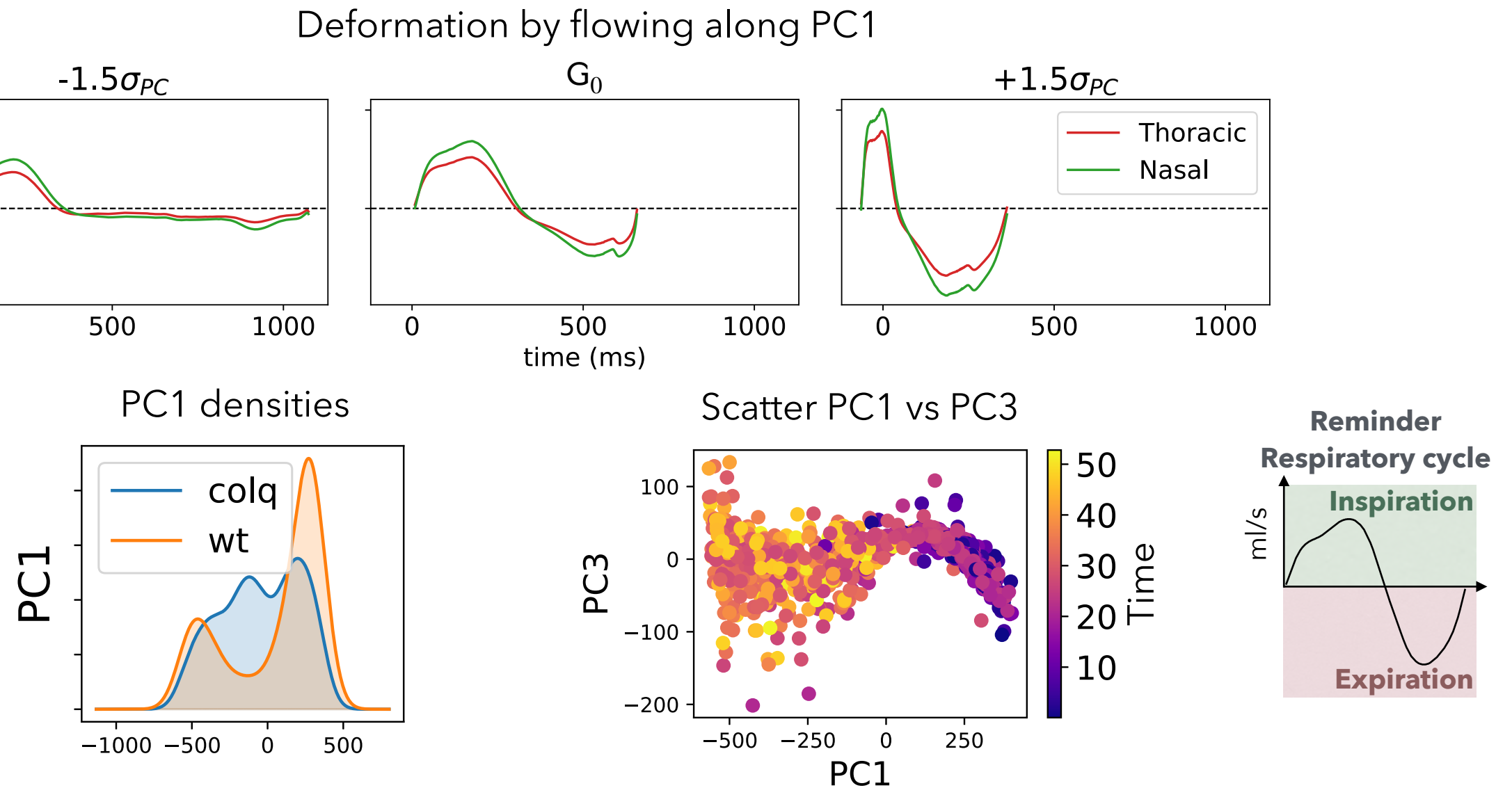
- Online learning of transfer operator ?
- Can we perform domain adaptation for policies based on transfer operator ?
- Implication in offline settings ? Dual representation ?

Thank you

Shape appendices

Whole exercise analysis

Kernel PCA applied on the initial velocity fields parametrized with $(\mathbf{G}_0, \alpha_j)_{j \in \llbracket 1, N \rrbracket}$.



Observations:

- PC1 negative tail indicates a pause between inspiration and expiration.
- WT mice are more affected than ColQ ones after drug injection.
- Diversity of responses to the drug injection (scatter plot).

Operator appendices

Ablation study

Protocol: Metric comparisons between a baseline linear dynamical system and a shifted one under 4 scenarios:

- (a) Frequency shift.
- (b) Decay shift.
- (c) Rank shift.
- (d) Sampling frequency shift.

